Teaching Philosophy Statement

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My primary goal as an instructor is to make sure every student feels welcome and respected, and knows they belong in the classroom. Students arrive with very different experiences of mathematics. I want to help them build confidence and find satisfaction in understanding something that once seemed difficult, and I am especially committed to supporting students from underrepresented and underprivileged communities. I share the convictions behind Federico Ardila-Mantilla's axioms for mathematics education: that mathematical talent is distributed equally across groups of people, and that every student deserves meaningful mathematical experiences and to be treated with dignity and respect.[1]

My own interest in mathematics took time to develop. As a child, I attended weekend classes taught by a physics teacher who traveled more than twenty miles to reach us. I struggled with mathematics and physics then, but I still vividly remember the experiments he showed us. As I solved more problems and began asking my own questions, I fell in love with mathematics. Those experiences remind me how much a teacher can do by making room for curiosity, and they are why I volunteer with math circles and with science and mathematics camps for schoolchildren.

Making Mistakes Is Part of Learning. When a student is unsure of an answer or makes a mistake, my first concern is to help them feel at ease. I remind the class that an incomplete idea is still worth discussing, and I openly acknowledge my own mistakes, too. When someone is stuck, I offer small hints and ask questions that help them find a next step. I want students to see revising an argument or trying another approach as part of doing mathematics, and to feel comfortable sharing their reasoning before they are certain it is correct.

Motivating New Ideas. To help students into an unfamiliar idea, I use geometric pictures and applications before introducing a formal definition. As instructor of record for MTH 314, Matrix Algebra with Applications, at Michigan State University (MSU) in Summer 2024, I used geometric transformations of the plane to motivate eigenvalues and eigenvectors. One student wrote that "we learned what these concepts actually represented in linear algebra rather than just the routine calculations." This captures a central goal of my teaching: students should be able to explain what their calculations mean. I will continue to build on this approach to help students connect mathematical ideas with their interests in mathematics, the sciences, and other fields.

Active Learning and Group Work. After introducing a concept and working through an example, I give students time to try a related worksheet problem before I present a solution. I circulate, offer small hints, and help students find a partner when they need one. While circulating, I monitor where their reasoning stalls. When several groups struggle with the same concept, I bring the class together at the board for a brief, targeted explanation before they resume their work. This uses class time efficiently and helps students recognize that their questions are shared by others. A classroom observer described the result as a "collaborative and productive classroom environment." I want participation to grow out of feeling included and having something to contribute.

In Spring 2025, I taught approximately sixty students as instructor of record for MTH 103A, College Algebra I. MTH 103A is the first half of a two-semester stretch sequence designed for students who benefit from additional time and structured in-class instruction compared with a

[1] Federico Ardila-Mantilla, *Todos Cuentan: Cultivating Diversity in Combinatorics*, *Notices of the AMS* **63** (2016), no. 10, 1164–1170.

standard one-semester course. This format allowed us to explore core algebraic concepts deeply while reinforcing prerequisite concepts as they arose. I prepared my own lecture notes and worksheets and worked with two undergraduate learning assistants. The course used standards-based grading on a pass/fail basis, with further opportunities to demonstrate mastery of topics students had initially found difficult. In my experience, this reduced students' stress: they had manageable goals, and an unsuccessful attempt became a reason to keep working rather than a final verdict. This experience reinforced my commitment to making expectations clear and giving students meaningful opportunities to act on feedback.

Supporting the Transition to Proof. As a recitation leader for MTH 317H, Honors Linear Algebra, I teach proof techniques that students need but may not yet know. I design worksheets that ask why a concept is useful or needed, and release complete solutions afterward so students can check their reasoning. These activities help students move from computation to working with definitions, testing conjectures against examples and counterexamples, and explaining why an argument works. Inspired by Professor Jinfa Cai's talk in my mathematics education class, I redesigned these worksheets in Fall 2026 around problem posing: students change hypotheses, formulate their own questions, and exchange problems with classmates. Students keep their worksheets, which include definitions and notes they can study from, and hand in only a short exit ticket at the end of each recitation. Since making these changes, I have seen student participation increase, and students have responded positively to the new format. The redesigned worksheets are available at s-adithya.github.io/teaching/mth317h-fs26.html. Building on this experience, I plan to design a mathematics education project in Spring 2027 to analyze carefully how effective this approach is in promoting active learning in MTH 299, Introduction to Proofs, and MTH 317H. In particular, I hope to understand how problem posing affects participation and the explanations students give for their reasoning.

Teaching with AI. Generative AI makes it especially important to design activities and assessments that reveal students' thinking. Where course policies permit, I would invite students to critique an AI-generated example or proposed solution, identify unsupported steps, and explain how to correct them. Students could also use AI to help prepare definition sheets or suggest examples to investigate, while checking mathematical claims against course materials. Any suggested references would need to be located and read in their original form. I would make expectations for acknowledging AI assistance explicit and provide equivalent activities that do not require AI access. Written reasoning, classroom discussion, and individual reflections would remain central to assessing understanding.

Mathematics Education Background. My coursework in *Teaching College Mathematics* gives me another way to reflect on these choices, and I am working toward the *Certification in College Teaching at MSU*. Discussions with classmates from several disciplines have encouraged me to consider offering students more choice in how they demonstrate understanding, through projects such as a video, artwork, or a physical model accompanied by an explanation of the mathematics involved. In developing such assignments, I would use shared mathematical criteria so that different formats serve the same learning goals.

Mentoring and Leadership. In Summer 2026, I mentored two undergraduates in the MSU Summer Topology Program, one on Euler characteristics and the Lefschetz fixed-point theorem and the other on the Gauss–Bonnet theorem and the classification of surfaces. When explaining chain complexes to a student who had not studied abstract algebra, I learned to start with familiar vector spaces and linear maps and use a simple example to motivate the general definition. This experience taught me to build a path into advanced ideas from what a student already knows. I have also spent

six semesters as a tutor and supervisor in the Mathematics Learning Center. As Lead Teaching Assistant in 2024–25, I observed and gave teaching feedback to eight teaching assistants and sixteen undergraduate learning assistants, strengthening my appreciation for collaborative reflection on teaching.

Future Teaching and Mentoring. Working closely with students, both in the classroom and through undergraduate research, is central to how I hope to teach and mentor. I would welcome opportunities to contribute across the undergraduate curriculum, including general education courses and courses for majors, drawing on my experience in college algebra, calculus, linear algebra, and proof-based instruction, and to develop advanced offerings connected to my interests in algebra and topology. To involve more students in undergraduate research, I would develop entry points through guided reading and exploratory projects, using concrete examples and regular conversations to help students move toward questions of their own. I would also welcome collaboration with department colleagues on innovative teaching methods, and on examples and projects that help students understand how mathematics supports their broader academic and professional goals.

Across these settings, from helping students build confidence in College Algebra and guiding proof-writing activities to supporting graduate algebra students and mentoring undergraduate topology projects, I keep returning to what that physics teacher showed me: students learn best when someone makes room for their curiosity and believes they can understand. Supported by my coursework in college teaching and experience as Lead Teaching Assistant, I will continue listening to student feedback and adapting my methods. I hope students leave my courses with mathematics they can use, and more confident asking questions, exploring ideas, and doing mathematics with others.

My teaching portfolio is available at s-adithya.github.io/teaching/portfolio.html.